

Final Project Proposal

Deep Learning for Computer Graphics

Shanying Liu

Material Aware Lighting Prediction From Single Images

Goal:

My main goal is to train a neural network to predict lighting parameters and material properties from a single image. This is in order to answer the question: does explicitly modeling material categories improve lighting estimation accuracy compared to approaches that ignore materials? If so, is this improvement uniform across all materials?

In other words, I will compare my multi-task model against a single-task baseline, and investigate lighting accuracy metrics for each material category.

Topics covered: utilizes multi-task learning, CNNs. Within Graphics this touches upon inverse rendering and the prediction of lighting and materials.

Background: I believe that this investigation has some novelty potential, since most work focuses on predicting lighting *or* materials separately, instead of evaluating whether improvements in light prediction can be made through material awareness.

Hypothesis:

I expect that explicitly modeling material categories will improve lighting prediction accuracy over the non-material aware lighting prediction, especially for materials with strong specular components due to pronounced surface reflectance. For diffuse materials I expect the improvement will be less noticeable.

Implementation plan:

Multi-task CNN:

I will use a ResNet-50 backbone as a feature extractor attached to two output heads. One head will predict lighting (27-dim spherical harmonics coefficients) with regression and MSE loss, while the other head predicts material class.

Create synthetic dataset:

- Use Blender + scripting to create dataset with distinct material categories (diffuse, glossy, metallic, rough metallic, maybe dielectric?) applied to spheres (for good representations of all normals) with randomly selected colors to avoid over-fitting.

- Also choose varying lighting directions/intensities. The lighting will come from free HDRIs found online, rotated at random angles, while the camera looks at only a sphere.
- Rendered images will likely be 256x256, but will start testing scripts on 128x128.
- Will save corresponding metadata about lighting and material category and parameters.

Training pipeline:

Create a training loop in PyTorch with a loss function combining lighting and material loss.

Evaluation:

- Test on held-out synthetic data and real photos.
- Compare lighting accuracy with and without material awareness.
- Lighting accuracy will be evaluated using angular error of light direction (from SH coefficients) and relative intensity error of the ambient and primary light components.
- Potentially also render spheres under the predicted vs ground truth lighting to compare images.

3rd Party APIs or frameworks:

PyTorch, Blender with python scripting for synthetic dataset, PyTorch 3d maybe?

Rough timeline:

- Week 1: Blender dataset generation (10k images). Collect HDRIs and write scripts.
- Weeks 2-4: Implement multi-task ResNet, training pipeline
- Week 5: Train and debug on synthetic data (Midpoint evaluation)
- Weeks 6: Evaluate on held-out synthetic test set
- Weeks 7: Capture real photos (20-30 objects) for qualitative testing
- Week 8: Final analysis, visualizations, presentation

Stretch goals:

- Add material parameter regression rather than material classification
- Test on real objects with complex shapes instead of only spheres

Resources:

Textbook: Prince, chapter 10.

Websites: [HDRI Source](#): [polyhaven](#)

Relevant papers:

- Neural Light Field Estimation for Street Scenes with Differentiable Virtual Object Insertion — original inspiration. Inserts virtual objects into outdoor street scenes (single image). Very complex, full pipeline for predicting light, creating HDR sky dome and volumetric gaussian grid, and rendering virtual objects into the image. Material-unaware, treats all as Lambertian.
- EMLight: Lighting Estimation via Spherical Distribution Approximation — Predicts SH coefficients from single images using a CNN (indoor focus). They regress spherical light distribution using a spherical mover's loss, but does not model scene materials. I would like to extend this to add material understanding.
- Neural LightRig: Unlocking Accurate Object Normal and Material Estimation with Multi-Light Diffusion — This uses one input image to estimate materials, which can be used to render relit images under different environment lighting, assuming light is given. I am essentially doing the opposite.
- On-site single image SVBRDF reconstruction with active planar lighting — uses a light-aware (planar light only) framework to reconstruct SVBRDFs from an image captured in uncontrolled environments